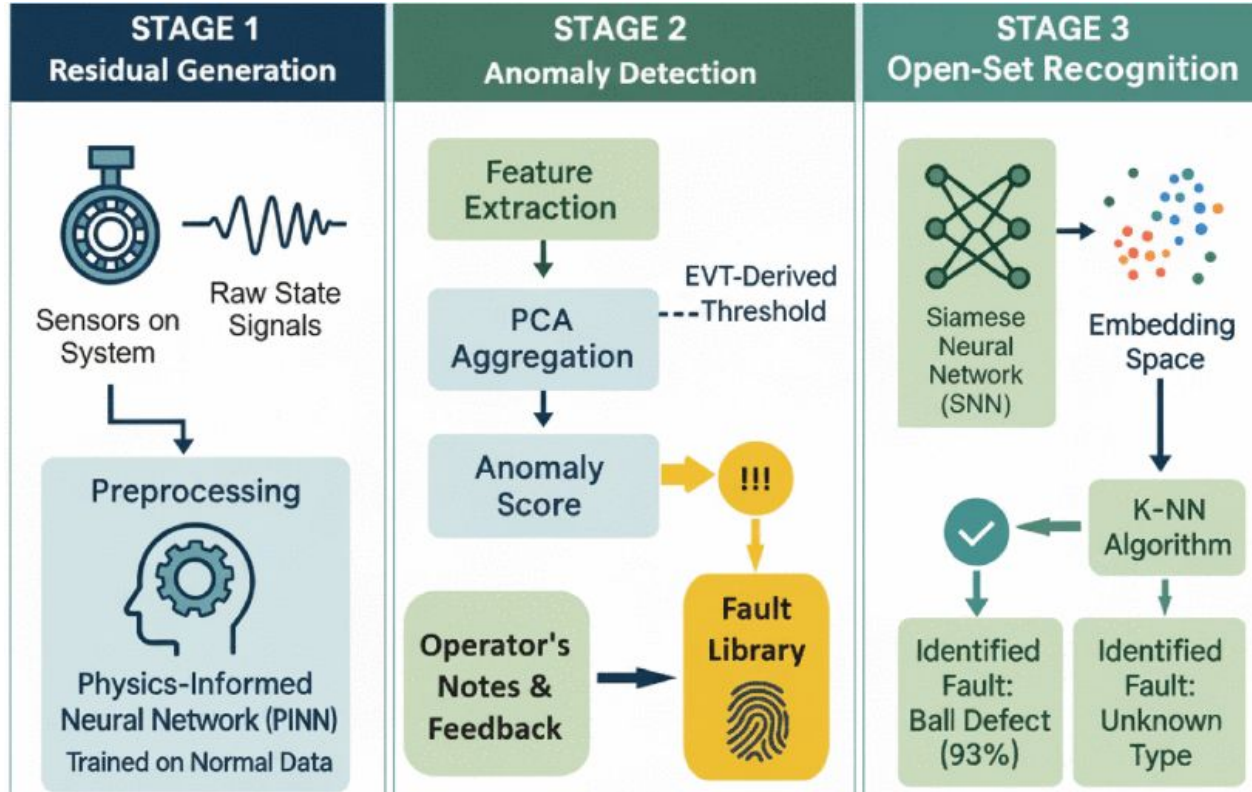


Physics-Guided Counterfactual Fault Synthesis in Cyber-Physical Systems

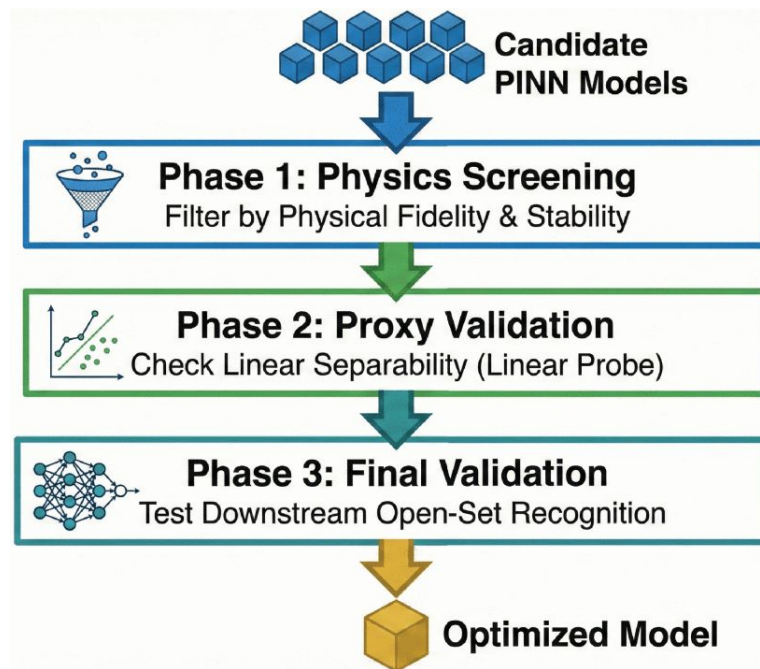
Presentation by Enzo Nicolás Spotorno, Josafat Ribeiro Leal and
Gustavo Rodrigues Alves D'Angelo

Software/Hardware Integration Lab, Federal University of Santa Catarina, Florianópolis, SC, Brazil

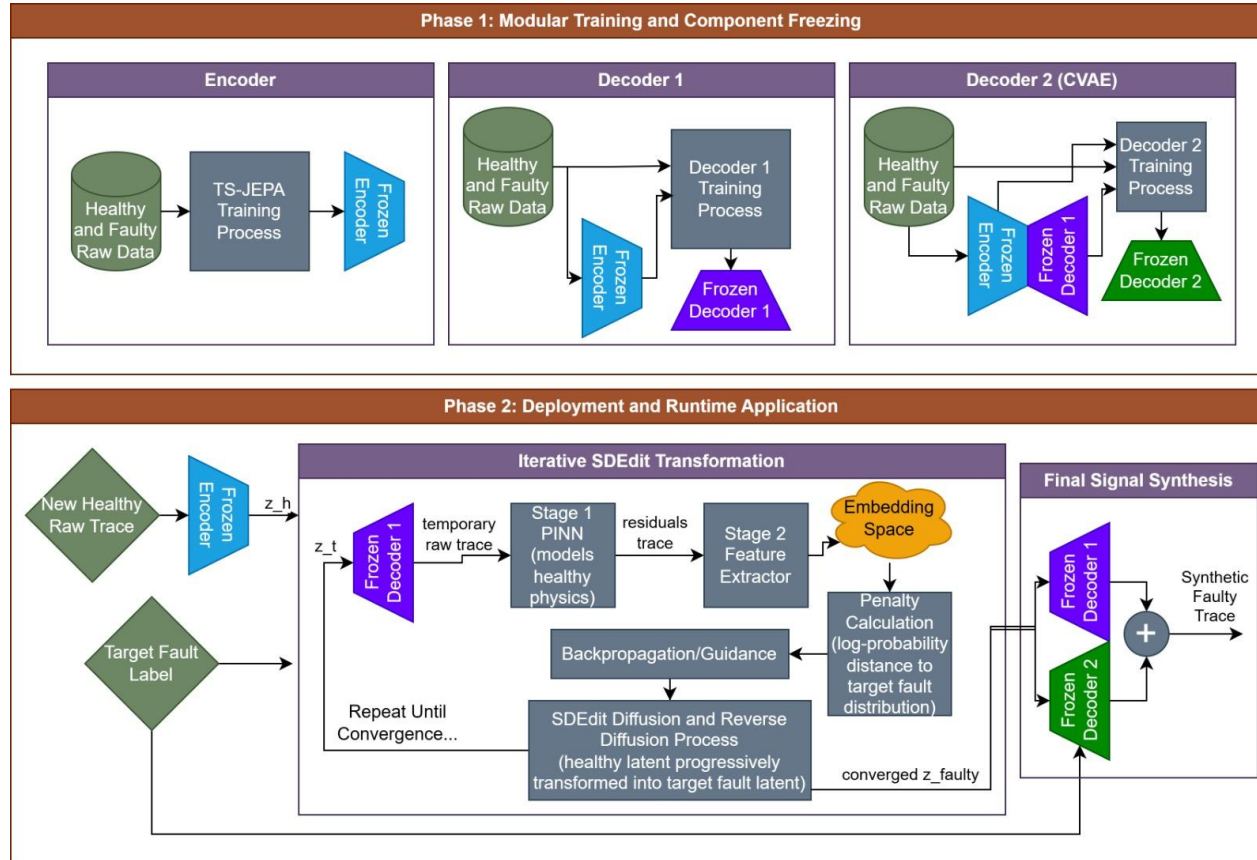
The Three-Stage Diagnostic Pipeline



Three-Phase Evaluation Protocol for Tractable Optimization



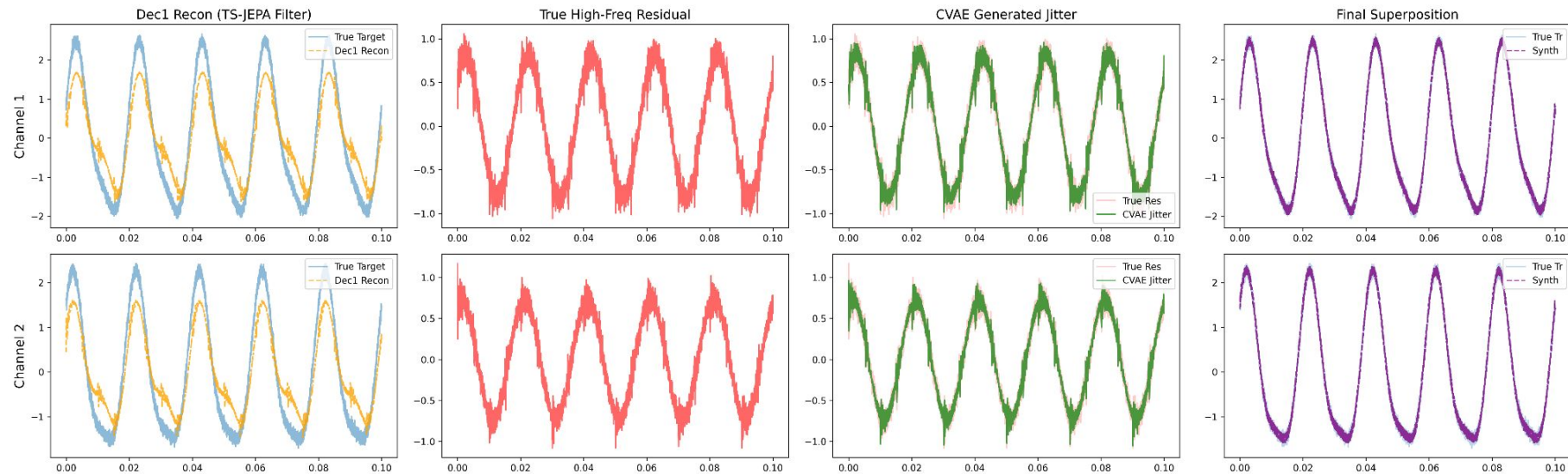
Methodology: Phases



First Proof of Concept: Evaluation Class 1



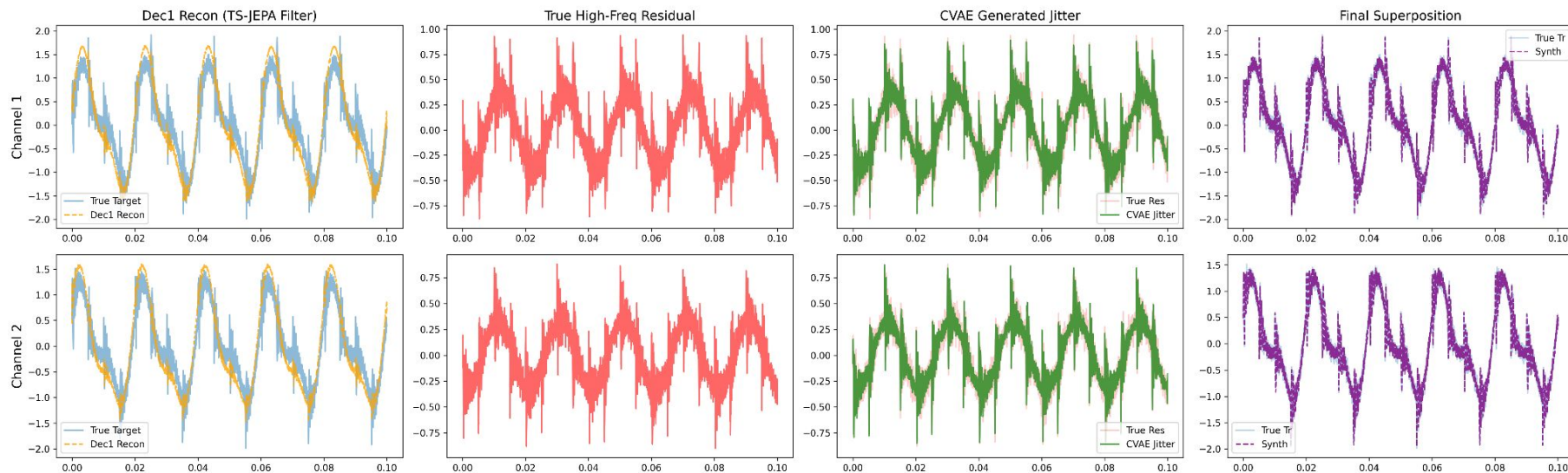
Multivariate Analysis: Imbalance Class



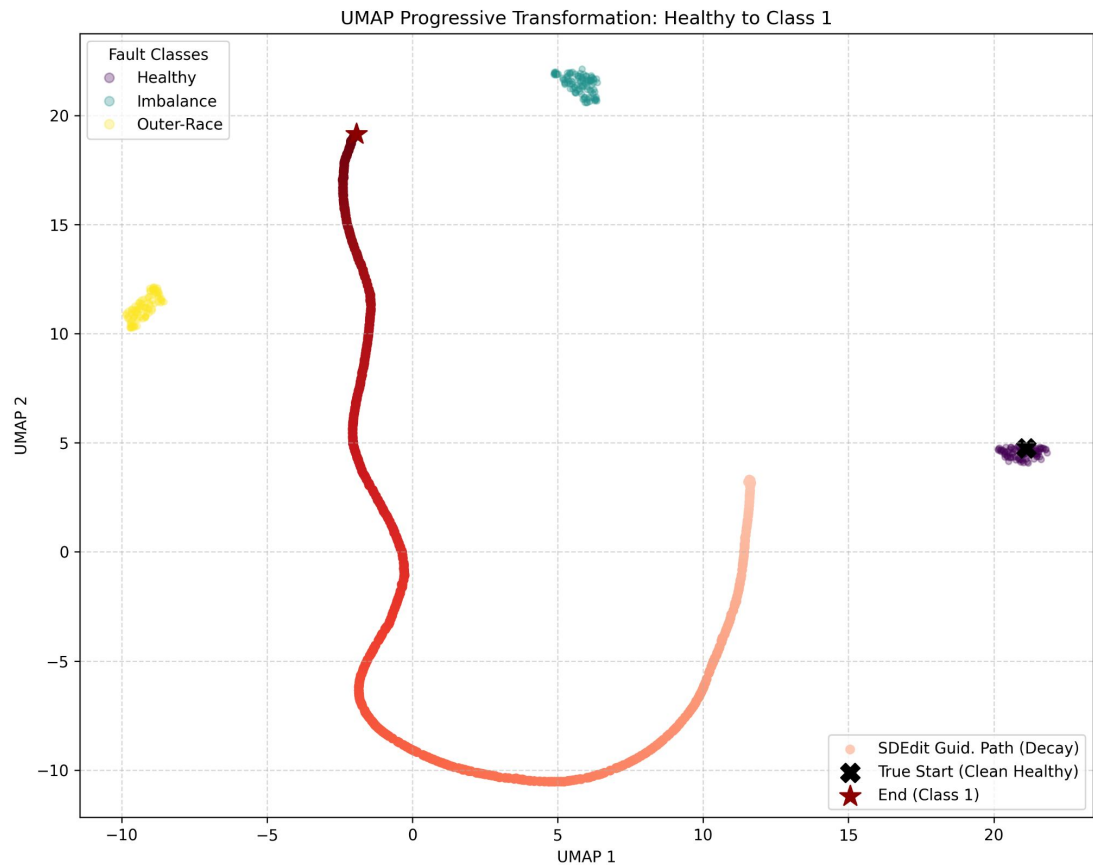
First Proof of Concept: Evaluation Class 2



Multivariate Analysis: Outer-Race Class



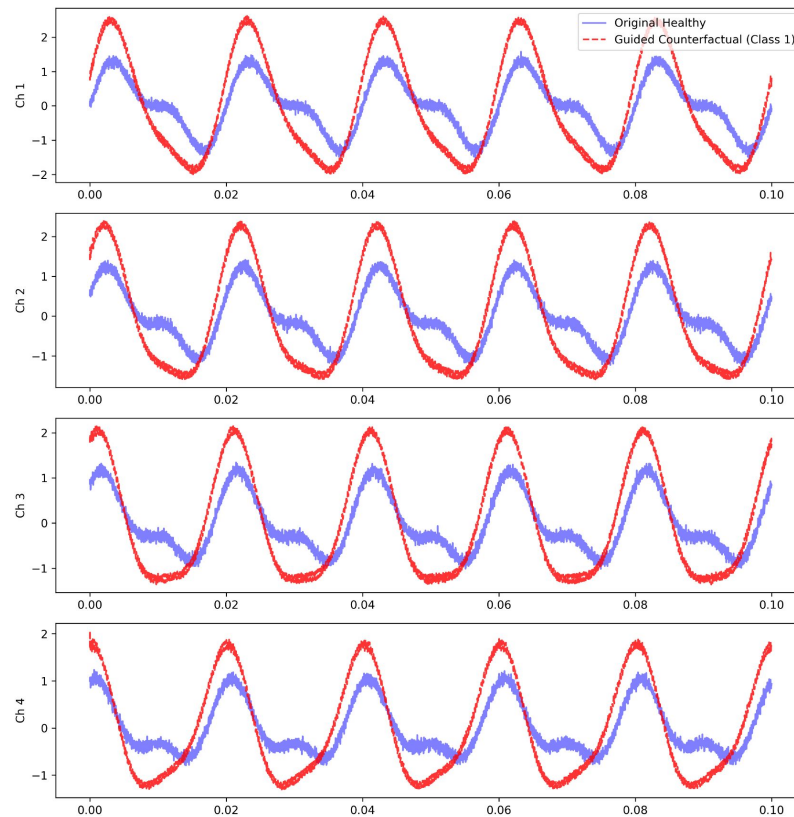
First Proof of Concept: Trajectory Class 1



First Proof of Concept:

Counterfactual Class 1

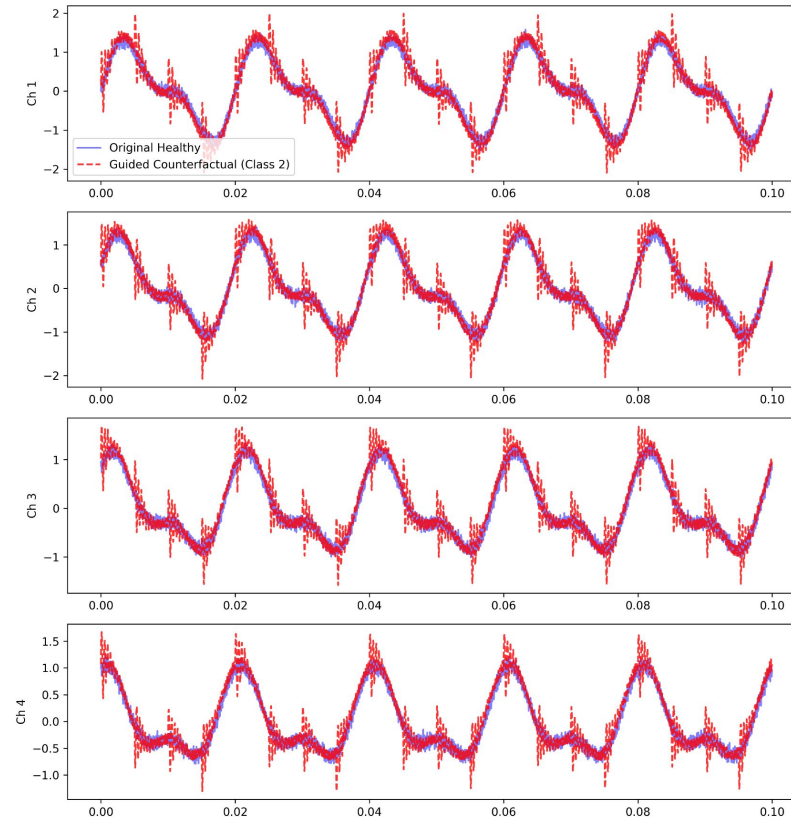
Phase 3: Physics-Guided Counterfactual Synthesis
Transition: Healthy -> Class 1



First Proof of Concept:

Counterfactual Class 2

Phase 3: Physics-Guided Counterfactual Synthesis
Transition: Healthy -> Class 2



Thanks!

Presentation by Enzo Spotorno
LISHA/UFSC